**10.** The curve $C_1$ has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (4x - 3)(x - 5)^2$$

(a) Sketch $C_1$ showing the coordinates of any point where the curve touches or crosses the coordinate axes.

**(3)**

(b) Hence or otherwise

(i) find the values of $x$ for which $\mathrm{f}\left(\frac{1}{4}x\right) = 0$

(ii) find the value of the constant $p$ such that the curve with equation $y = \mathrm{f}(x) + p$ passes through the origin.

**(2)**

A second curve $C_2$ has equation $y = \mathrm{g}(x)$, where $\mathrm{g}(x) = \mathrm{f}(x + 1)$

(c) (i) Find, in simplest form, $\mathrm{g}(x)$. You may leave your answer in a factorised form.

(ii) Hence, or otherwise, find the $y$ intercept of curve $C_2$

**(3)**